

SCIENCE AND SCIENTIFIC PROGRESS

I) SPACE EXPLORATION

1) Definition

Investigation, by means of crewed and uncrewed spacecraft, of the reaches of the universe beyond Earth's atmosphere and the use of the information so gained to increase knowledge of the cosmos and benefit humanity.

2) The purpose of space exploration

Humans have always looked at the heavens and wondered about the nature of the objects seen in the night sky. With the development of rockets and the advances in electronics and other technologies in the 20th century, it became possible to send machines and animals and then people above Earth's atmosphere into outer space. Well before technology made these achievements possible, however, space exploration had already captured the minds of many people, not only aircraft pilots and scientists but also writers and artists.

Achieving spaceflight enabled humans to begin to explore the solar system and the rest of the universe, to understand the many objects and phenomena that are better observed from a space perspective, and to use for human benefit the resources and attributes of the space environment. All of these activities—discovery, scientific understanding, and the application of that understanding to serve human purposes—are elements of space exploration.

3) Human beings in space: debate and consequences

By the early 2020s more than 500 people, coming from more than 40 different countries and more than 10 percent of whom were women, had flown in space. As of that same time, only Russia, China, and the United States had the capability of carrying out human spaceflights. With the retirement of the space shuttle in 2011, the United States lost its independent human spaceflight capability. Such capability was not regained until 2020, when a new private commercial spacecraft, SpaceX's Crew Dragon, was ready for use.

4) Commercial space transportation

The prosperity of the communications satellite business was accompanied by a willingness of the private sector to pay substantial sums for the launch of its satellites.

However, in the 2010s, SpaceX's partially reusable Falcon 9 rocket came to dominate the commercial launch industry with its low prices. The air force awarded a contract to SpaceX. In 2018, out of 41 commercial launches, the Falcon 9 did 16.

In 2017 the revenues of the commercial space transportation industry were estimated to be \$5.5 billion. Analysts forecast an average of 42 commercial launches per year in the ensuing decade.

5) New commercial applications

Access to low Earth orbit has typically cost tens of thousands of dollars per kilogram of payload—a significant barrier to further space development. However, one company, SpaceX, lowered this cost by a factor of 10 with its Falcon 9 rocket and promises to reduce it still further with its planned Falcon Heavy.

Another potential commercial application is the transport of fare-paying passengers into space, known as space tourism. Various surveys have suggested a willingness among many in the general public to spend considerable sums for the opportunity to experience space travel. Although a very limited number of wealthy individuals have purchased trips into Earth orbit to visit the ISS at a very high price, large-scale development of the space tourism market will not be possible until less-expensive, highly reliable transportation systems to orbit have been developed.

Resources available on the Moon and other bodies of the solar system, particularly asteroids, represent additional potential objectives for commercial development. For example, over billions of years the solar wind has deposited large amounts of the isotope helium-3 in the soil of the lunar surface.

Scientists and engineers have suggested that helium-3 could be extracted and transported to Earth, where it is rare, for use in nuclear fusion reactors. In addition, there is evidence to suggest that the Moon's polar regions contain ice, which could supply a crewed lunar outpost with drinking water, breathable oxygen, and hydrogen for spacecraft fuel. Significant quantities of potentially valuable resources such as water, carbon, nitrogen, and rare metals may also exist on some asteroids, and space mining of those resources has been proposed.

II) GENETIC ENGINEERING

1) Definition

The artificial manipulation, modification and recombination of DNA or other nucleic acid molecule in order to modify an organism or population of organisms.

2) Historical developments

Initially referred to various techniques used for the modification or manipulation of organisms through the processes of heredity and reproduction ® artificial insemination, in vitro fertilization (IVF, "test tube" babies), cloning and gene manipulation

Late 20th century: recombinant DNA technology (gene cloning): DNA molecules from two or more sources are combined either with cells or in vitro and are inserted into host organisms in which they are able to propagate

3) Gene editing

Definition: the ability to make highly specific changes in the DNA sequence of a living organism, essentially customising its genetic make up

Gene therapy: the introduction of a normal gene into an individual's genome in order to repair a mutation that causes a genetic disease

4) Controversies and ethical issues

Patents on genetically engineered and genetically modified organisms, particularly crops and other foods

Fear that it might result in the introduction of unfavourable and possibly dangerous traits into microorganisms that were previously free of them: resistance to antibiotics, production of toxins etc. The impact of genetically modified crops on human and environmental health. Ex: genetic manipulation may alter the allergenic properties of crops

Ethical and social implications on the genetic engineering of humans: its potential use to alter traits such as intelligence or beauty, alter genes in human sperm ® those genes could be passed on to subsequent generations, or increase the risk of cancer

III) CRISPR: Clustered Regularly Interspace Short Palindromic Repeats

1) How it works

A sort of genetic memory enabling the cells to efficiently detect and destroy pathogens, particularly viruses known as bacteriophages.

Allows researchers to remove and insert DNA in desired locations within a genome, making it possible to correct genetic defects in animals and to modify DNA sequences in embryonic stem cells.

Discovered by Jennifer Doudna and Emmanuelle Charpentier in 2012 ® Nobel Prize in Chemistry in 2020.

Applications: clinical trials for the treatment of certain human diseases ® 2023: first CRISPR-based treatment approved by the Food and Drug Administration (FDA) for patients affected by sickle-cell disease.

2) Ethical considerations

Modifying the germ line (i.e. sperm / eggs) may lead to designer babies: improve traits like intelligence, appearance, etc. ® Is it moral to temper with human development and who would have access to the technology?

2018: twin girls carried an genetic change that reduces the risk of HIV infection in China ® moratorium on editing human genes in eggs, sperm or embryos: scientists may unintentionally introduce as many genetic defects as they attempt to fix.

IV) **BIOETHICS**

A) Definition

Branch of applied ethics that studies the philosophical, social and legal issues arising in medicine and life sciences.

B) The issues in bioethics

1) The relationship between doctor and patient

Conflict between a doctor's duty to promote the health of their patient and the patient's right to self-determination or autonomy. Ex: terminally-ill patients, those who cannot make rational decisions in their own interest.

2) Traditional philosophical questions: the definition of life and death, the nature of personhood and identity, and the extent of human freedom and responsibility

When is a fatally injured or terminally ill patient considered dead?

The moral status of the human foetus ® is the foetus a person in the moral sense? At what stage? ® Abortive pill + use of embryonic stem cells raised the issue

Cloning: what makes a particular human being to unique person they are? Are they just the sum of information encoded in their genes? ® Gene therapy: how do you become someone else? Human cloning: would they be a copy of the "parent"? Will the two be the same person?

Free will and determinism: to what extent is human personality or character genetically rather than environmentally determined?

Technologies that would extend the human lifespan: speculation about the value of life, the significance of death, and the desirability of immortality. What would the significance of death be a the world in which dying was not biologically inevitable?

3) Social and legal issues

Disproportionate disadvantages for some social groups, either by being applied in ways that harm members of the group strictly or by encouraging the adoption of social qualities that discriminate unfairly against them ® reproductive medicine: choose the sex of the baby, abortion used as a method of birth control in China and India (boys more valued than girls).

Genetic tests to determine a patient's susceptibility to certain genetically transmitted diseases could be used in unfair ways by health insurance companies, employers and government agencies.

Gene counselling: people have become less tolerant of disability

4) Research on the genetic bases of behaviour

Encourage the adoption of crude models of genetic determinism in the development of social policies, especially in the areas of education and crime prevention ® unfair discrimination against large numbers of people judged to be genetically disposed to "undesirable" forms of behaviour, such as aggression or violence.

5) Voluntary euthanasia

Morally permissible in some circumstances does not settle by itself the question of whether euthanasia should be legalised ® potential for abuse. Ex: elderly patients could think that society expects them to request euthanasia so as not to be a burden to others.

6) The allocation of healthcare resources
An individual or a public responsibility?

7) The four principles approach
Autonomy, beneficence, nonmaleficence, justice

V) THE PRONATALIST MOVEMENT

1) Definition

Political, ideological, or religious projects to encourage childbearing by some or all members of a central, ethnic or national group (cf. Elon Musk has 11 children)

2) The movement

Malcolm and Simone Collins, the leading spokespeople of pronatalism, screen the embryos for disability and optimise for intelligence.

The child is less an individual, with desires and dignity, than a vehicle for a political project ® fear of the white population being replaced by non-white immigrants.

VI) DE-EXTINCTION

1) Definition

De-extinction or resurrection biology, reverses plant and animal extinctions by creating new versions of previously lost species ® re-establish dynamic processes that produce healthy ecosystems and restore biodiversity. **IT IS NOT POSSIBLE TO RESURRECT EXTINCT SPECIES IN THEIR GENETIC, BEHAVIOURAL AND PHYSICAL ENTIRETY.**

2) The ways de-extinction works

Back-breeding uses the principles of selective breeding ® increases the presence of specific traits within a population

Cloning requires intact living cells ® more suitable for population where some individuals remain

Gene editing: Gene editing tools delete or add small sections of DNA in the genome ® change in physical traits (ex: eye colour)

Synthetic genomics: scientists usually attach long sections of DNA and introduce them into biological entity's genome ® novel genes or genes found in other organisms

3) How does de-extinction work through genome editing?

It creates hybrids between living and extinct organisms ® insert genes from an extinct species into the nucleus of a reproductive cell.

In April 2025, Colossal Biosciences announced the birth of three healthy dire wolf pups.

